

COURSE TITLE : HEAT TRANSFER & EVAPORATION
COURSE CODE : 4072
COURSE CATEGORY : B
PERIODS/ WEEK : 5
PERIODS/ SEMESTER : 70
CREDIT : 5

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Conduction, Fourier law, variation of thermal conductivity with temperature	18
2	Convection-natural and forced convection, significance of heat transfer coefficients	17
3	Radiation-Kirchoff's law, radiation between two surfaces.	18
4	Heat transfer equipments, Multiple effect evaporator.	17
TOTAL		70

COURSE OUT COMES :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	Understand the mechanism of heat transfer by conduction.
	2	Calculate the amount of heat loss through flat composite wall, cylinder and sphere.
2	3	Understand the mechanism of heat transfer by convection.
	4	Study film coefficient, overall heat transfer coefficients.
	5	To solve the problems relating radiation mode of heat transfer.
3	6	Compare and study construction, working and application of various types of heat transfer equipments.
	7	Analyze an evaporator problem and to calculate the material and energy requirement of an evaporator.
	8	Understand the general principles underlying multiple effect evaporator.

SPECIFIC OUTCOMES

MODULE – I

1.1.0 Understand the mechanism of transfer of heat by conduction

- 1.1.1 Define conduction.
- 1.1.2 State and explain Fourier's law of thermal conduction.
- 1.1.3 Apply rate equation for heat flow.
- 1.1.4 Derive the conduction equation for plane wall.
- 1.1.5 Define thermal conductivity.
- 1.1.6 Solve the simple problem apply the basic equation of thermal conduction.
- 1.1.7 Explain the variation of thermal conductivity with the help of solids, liquids, gases.

- 1.1.8 Distinguish between steady state and unsteady state conduction.
- 1.1.9 Derive the equation to calculate heat transfer through composite plane wall.
- 1.1.10 Derive the equation to calculate heat transfer through cylindrical wall.
- 1.1.11 Derive the equation to calculate heat transfer through spherical wall.
- 1.1.12 Solve the problems using above derived equations.
- 1.1.13 Explain the economic thickness of insulation.
- 1.1.14 Explain the different type of insulating materials.
- 1.1.15 Explain the characteristic of good insulating materials.

1.2.0 Comprehend the transfer of heat by Convection.

- 1.2.1 Define and explain the convection.
- 1.2.2 Explain the film concept in heat convection.
- 1.2.3 Draw and explain the temperature gradient in forced convection.
- 1.2.4 Define and explain individual heat transfer coefficient.
- 1.2.5 Explain the factors affecting the heat transfer coefficient.
- 1.2.6 Compute the values of heat transfer coefficients using empirical equation
- 1.2.7 Explain the factors influencing heat transfer coefficient in boiling liquid and condensing vapours.
- 1.2.8 Explain the concept of overall heat transfer coefficients based on inside and outside area.
- 1.2.9 Derive an equation for overall heat transfer coefficient from film coefficients.
- 1.2.10 Solve the problems using the equation derived.

MODULE – II

2.1.0 Understand the mechanism of Natural and Forced convection

- 2.1.1 Explain the term forced convection.
- 2.1.2 Distinguish between natural convection and forced convection.
- 2.1.3 Give the examples to forced convection and natural convection.

2.2.0 Understand transfer of heat by free convection.

- 2.2.1 Explain the mechanism of natural convection.
- 2.2.2 Explain the dimensionless equations.
- 2.2.3 Define Nusselt number.
- 2.2.4 Define Reynold number.
- 2.2.5 Prandtl number.
- 2.2.6 Define Grashof number.
- 2.2.7 Explain the relation between Nusselt number and Reynold number and Prandtl number.
- 2.2.8 Define Dittus Boilers equation and sieder-Tate equations.
- 2.2.9 Estimate the heat transfer coefficient from above equation.
- 2.2.10 Solve the problems relating these equations.

2.3.0 Comprehend heat transfer in boiling liquid and condensing vapour.

- 2.3.1 Explain heat transfer to boiling liquid.
- 2.3.2 Define pool boiling of saturated liquid.
- 2.3.3 Define sub cooled boiling.

- 2.3.4 Explain pool boiling of saturated liquid with figure ,heat flux Vs temperature drop.
- 2.3.5 Define Leiden frost point.
- 2.3.6 Define nucleate boiling.
- 2.3.7 Define transition boiling.

2.4.0 Know the heat transfer in fluid with phase change

- 2.4.1 Explain heat transfer from condensing vapour.
- 2.4.2 Explain film type condensation.
- 2.4.3 Explain Drop wise condensation.

2.5.0 Understand radiation as a mode of heat transfer.

- 2.5.1 Explain Radiation.
- 2.5.2 Explain the concept of black body, white body, gray body.
- 2.5.3 Explain the laws of Radiation, Krichoff's, Plank's law, Stefan Boltzman, wien's law.
- 2.5.4 Solve the problem relating above laws .
- 2.5.5 Explain radiation heat transfer between two solids.

MODULE – III

3.1.0 Understand the construction and working of heat transfer Equipments

- 3.1.1 Distinguish between parallel flow, counter current flow and cross flow in heat exchangers.
- 3.1.2 Explain the term fouling factors in heat exchangers.
- 3.1.3 Fouling effect calculations in heat exchangers.
- 3.1.4 Explain the concept of varying temperature drop and log mean temperature drop.
- 3.1.5 Solve the problem using log mean temperature drop equation in counter flow, parallel flow and cross flow
- 3.1.6 Compute the heat transfer area of heat exchanger.
- 3.1.7 Estimate the tube length of a heat exchanger.
- 3.1.8 Explain general features of shell and tube heat exchanger.
- 3.1.9 Explain the constructional details and working of a single pass shell and tube heat exchanger with the help of sketch.
- 3.1.10 Explain the concept of multi passes in heat exchangers and economic passes.
- 3.1.11 Explain the constructional details and working of U tube heat exchangers.
- 3.1.12 Explain the constructional details of floating head type heat exchangers.
- 3.1.13 Explain the functions of baffles on shell side of a heat exchanger.
- 3.1.14 Explain the constructional details and working of double pipe heat exchanger
- 3.1.15 Explain the constructional details and working of plate type heaters, air fin cooler and finned tubes.
- 3.1.16 Explain the simple principles of installation and maintenance of heat exchangers

MODULE – IV

4.1.0 Understand the construction and working of evaporators

- 4.1.1 List the application of evaporation operation in chemical industries.
- 4.1.2 Classify evaporators.
- 4.1.3 Explain the construction and working of horizontal tube evaporator.
- 4.1.4 Explain the construction and working of short tube evaporator with help of sketch.
- 4.1.5 Explain the construction of and working of Basket type evaporator with the help of sketch.
- 4.1.6 Explain the construction and working of falling film type evaporator with the help of sketch.
- 4.1.7 List the advantage and disadvantage of above evaporators.
- 4.1.8 List the application of each of those evaporator
- 4.1.9 List the materials of construction of tubes and evaporator body
- 4.1.10 Explain the principle underlying the selection of evaporator for specific requirement.
- 4.1.11 Explain the control of evaporator operation.
- 4.1.12 Explain the special type of evaporators.
- 4.1.13 Explain the operational difficulties in evaporators.
- 4.1.14 Explain construction and working of different condensers.
- 4.1.15 Explain the working of vacuum pump
- 4.1.16 Explain the condensate removing equipments different type of traps.
- 4.1.17 Explain operating temperature range.
- 4.1.18 Explain the use of vacuum in evaporators.
- 4.1.19 Explain the concept of foam and entrainment.
- 4.1.20 Explain the method of air removal and salt removal.
- 4.1.21 Explain the scale formation in evaporators and its removal methods.

4.2.0 Compute the material balance and energy balance in single effect evaporator

- 4.2.1 Describe the material balance and heat balance equation in single effect evaporator.
- 4.2.2 Drive the equation to calculate the heat transfer area, steam requirement, capacity and steam economy.
- 4.2.3 Define the term capacity of an evaporator.
- 4.2.4 Define the term economy of an evaporator.
- 4.2.5 Explain boiling point elevation.
- 4.2.6 State Duhring's rule.
- 4.2.7 Draw the Duhring's line for NaCl-water.
- 4.2.8 Solve the problem relating to area, capacity and economy using Duhring's line
- 4.2.9 Explain the thermal effect of boiling point elevation
- 4.2.10 Explain the effect of Hydro static head

4.3.0 Understand the general principles underlying multiple effect evaporation

- 4.3.1 Explain the principles of multiple effect evaporation.
- 4.3.2 Draw the layout of a multiple effect evaporation system.
- 4.3.3 Explain the four methods of feeding in multiple effect system.
- 4.3.4 Explain the advantage and disadvantage of each system.
- 4.3.5 Explain the temperature distribution in multiple effect system with special Reference to boiling point elevation

- 4.3.6 Solve the problems to calculate the concentration of liquor in each effect and temperature in each effect

4.4.0 Understand the principle of vapour compression evaporation

- 4.4.1 Explain the principle of mechanical and thermo compression
4.4.2 Explain the industrial use of vapour compression
4.4.3 Explain the advantage of vapour compression

COURSE CONTENT

MODULE – I

Heat Transfer by Conduction

Heat transfer by conduction in solids – steady state and unsteady state flow – definition units of heat flow. Fourier's law of conduction – Rate equation for heat flow – steady state heat flow conduction through single wall – derivation of equation and simple problems – Thermal conductivity – units. Steady state conduction through composite wall in series derivation of equation and problems-Steady state conduction through cylindrical wall and spherical wall derivation – problems Theory of convection – film concept of heat transfer temperature gradient in forced convection – derivation of overall heat transfer coefficient from individual heat transfer coefficient – simple problems

MODULE – II

Forced Convection and Radiation

Forced convection: Heat transfer by forced convection inside tubes for laminar and turbulent flow – heat transfer to fluid without phase change – Dimensionless equation-Reynolds number - Nusselt number -Prandtl number- Grashof number - Film coefficients in pipes laminar flow - sieder -Tate equation Film coefficient in turbulent flow - Dittus Boelters Equation – simple problems – mechanism of natural convection – heat transfer in boiling liquids – flash boiling sub cooled boiling – saturated boiling – regimes of boiling – maximum and minimum radiation – elementary idea of black body – gray body- emissivity –emissive power – radiation laws. Stefan Boltzman equation – simple problems – radiation between surfaces – simple problems – radiation error in pyrometry – combined heat losses by convection and radiation.

MODULE – III

Heat Transfer Equipments

Parallel flow – counter current flow – cross flow heat exchangers – Fouling effect calculation – LMTD – calculation in heat exchangers – Heat transfer equipment – Heaters and heat exchangers – single pass shell and tube heaters Multipass heaters – economic number of passes, allowances for expansion in heaters and heat exchangers – Fixed tube sheet heat exchanger- U tube heat exchanger -floating head heaters - Kettle type reboilers – use of baffles on shell side of heat exchangers- double pipe heat exchangers – plate type heat exchangers – Fin type heat exchangers– calculation of heat transfer area and length of tubes – solve problems.

MODULE – IV

Constructional Details and Working Of Evaporator

Evaporation – Examples of industries where evaporation is used as a unit operation. Types of evaporators– basis of classification – horizontal tube – vertical tube – climbing film – falling film – forced circulation evaporators – examples of application of each in industries - continuous operation and control of evaporators – evaporator accessories – entrainment separator – condensers, vacuum pump – steam trap – salt catchers – continuous salt removal, evaporation under vacuum – operational difficulties.

Material and heat balance equation for single effect evaporators – simple problems – calculation of heating area – capacity – steam economy – factors that control heat transfer coefficient and effect of hydro static head – boiling point elevation.

Multiple effect evaporation – principle of multiple effect evaporation – operation of multiple system –method of feeding – advantages of multiple effect evaporation – steam economy and capacity. Temperature distribution in multiple effect and effect of boiling point elevation – optimum number of effect – simple problems in determining concentration and temperature in each effect Vapour compression evaporation –mechanical and thermo compression basic principles with specific industrial uses .

REFERENCE

Unit Operation – I	-	T.T.T.I. Kalamassery
Introduction to Chemical Engineering	-	Water.L.Badger
Unit operation of Chemical Engineering	-	Warren.L.McCabe and
Heat transfer	-	D.D.Tiwari
Fundamentals of Engg., Heat and Mass	-	R.C.Sachdeva
Heat and Mass Transfer	-	R.K.Rajput